

ECO Chlorophyll Fluorometer Characterization Sheet

Date: 4/2/2026

S/N: BBFL2WB-1307

Chlorophyll concentration expressed in $\mu\text{g/l}$ can be derived using the equation:

$$\text{CHL } (\mu\text{g/l}) = \text{Scale Factor} * (\text{Output} - \text{Dark counts})$$

	Digital
Dark counts	56 counts
Scale Factor (SF)	0.0182 $\mu\text{g/l/count}$
Maximum Output	4130 counts
Resolution	1.2 counts
Ambient temperature during characterization	22.0 °C

Dark Counts: Signal output of the meter in clean water with black tape over detector.

SF: Determined using the following equation: $\text{SF} = x \div (\text{output} - \text{dark counts})$, where x is the concentration of the solution used during instrument characterization. SF is used to derive instrument output concentration from the raw signal output of the fluorometer.

Maximum Output: Maximum signal output the fluorometer is capable of.

Resolution: Standard deviation of 1 minute of collected data.

The relationship between fluorescence and chlorophyll-a concentrations in-situ is highly variable. The scale factor listed on this document was determined using a mono-culture of phytoplankton (*Thalassiosira weissflogii*). The population was assumed to be reasonably healthy and the concentration was determined by using the absorption method. To accurately determine chlorophyll concentration using a fluorometer, you must perform secondary measurements on the populations of interest. This is typically done using extraction-based measurement techniques on discrete samples. For additional information on determining chlorophyll concentration see "Standard Methods for the Examination of Water and Wastewater" part 10200 H, published jointly by the American Public Health Association, American Water Works Association, and the Water Environment Federation.

ECO CDOM Fluorometer Characterization Sheet

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CDOM concentration expressed in ppb can be derived using the equation:

$$\text{CDOM (ppb)} = \text{Scale Factor} * (\text{Output} - \text{Dark Counts})$$

Dark Counts	Digital
Scale Factor (SF)	56 counts
Maximum Output	0.0905 ppb/count
Resolution	4155 counts
	1.2 counts
Ambient temperature during characterization	22.0 °C

Dark Counts: Signal output of the meter in clean water with black tape over detector.

SF: Determined using the following equation: $SF = x \div (\text{output} - \text{dark counts})$, where x is the concentration of the solution used during instrument characterization. SF is used to derive instrument output concentration from the raw signal output of the fluorometer.

Maximum Output: Maximum signal output the fluorometer is capable of.

Resolution: Standard deviation of 1 minute of collected data.

NTU Characterization Sheet

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Nephelometric Turbidity Unit (NTU) Scale Factor

Turbidity units expressed in NTU can be derived using the equation:

$$\text{NTU} = \text{Scale Factor} \times (\text{Output} - \text{Dark Counts})$$

			Digital
Dark Counts		50	counts
NTU Solution Value		2131	counts
Scale Factor (SF)		0.0495	NTU/count
Maximum Output		4130	counts
Resolution		1.0	counts
Ambient temperature during calibration	22	°C	

Dark Counts: Signal output of the meter in clean water with black tape over detector.

SF: Scale factor is determined using the following equation: $\text{SF} = \text{xx} \div (\text{Output} - \text{Dark Counts})$, where xx is the value of a Formazin concentration. For example: $12.2 \div (2011 - 50) = 0.0062$.

Maximum Output: Maximum signal output the meter is capable of.

Resolution: standard deviation of 1 minute of collected data.